Saint-Petersburg school of Economics and Management

International Program in Business and Economics

Data-Driven Company

Developing the forecast system of fuel consumption for the bus fleet considering routing network

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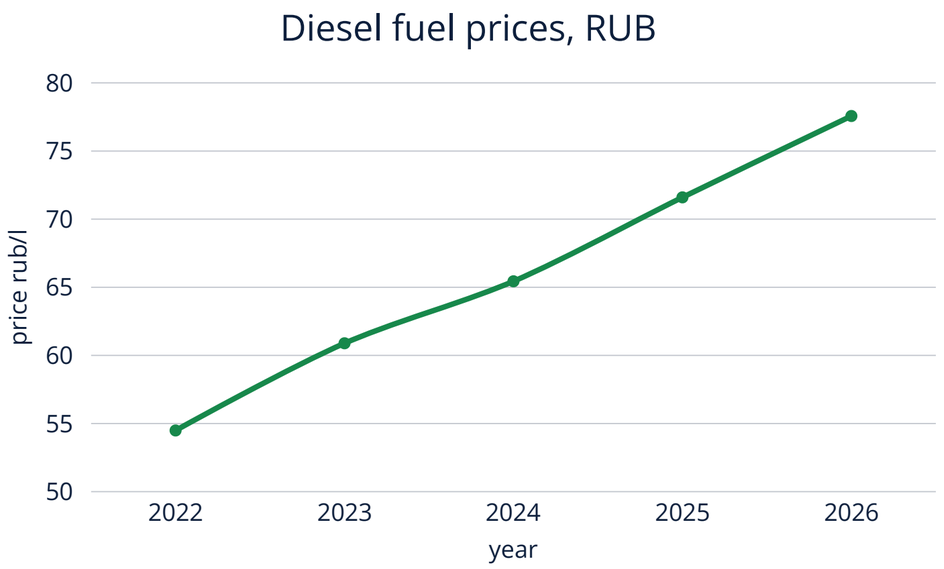
St. Petersburg, 2026

INDUSTRY CONTEXT

DIESEL FUEL

29 %
diesel price increase 2022–2025

30 %
share of fuel in operating expenses



(Source: compiled by the authors based on open statistical data)

FIXED-TARIFF GOVERNMENT CONTRACTS

- *Carriers bid a fixed price for contracts*
- *All cost overruns are absorbed by the carrier*

TWO COSTLY OUTCOMES

UNDERBID

Fuel costs exceed budget

direct, irreversible financial loss on contract

OVERBID

Higher bid

tender lost to a cheaper competitor; revenue gap

COMPANY CONTEXT

LLC «Palmira» - a major licensed transportation company servicing the urban and suburban route network of the Leningrad Oblast.

300+

***buses** with GLONASS*

20

***years** in the transportation market*

150+

***routes** across 7 districts of Leningrad Oblast*



Estimates fuel costs using Ministry of Transport normative (Order AM-23-r)

+

Actual fuel consumption depends on various factors

=

Palmira cannot accurately predict fuel consumption when submitting applications for state tenders

Research Question

What system **requirements, data inputs, and analytical models** are needed to build a **fuel consumption prediction system** for Palmira’s bus fleet that supports procurement planning and government contract bidding?

Literature review

How do organisations realise value from telematics infrastructure?

Telematics generates value only when embedded in **decisions**, not used for compliance alone

Powell & Chandran (2018)

Fuel management = largest gap between **potential** and **realised** telematics value across road transport

Miler et al. (2020)

Organisational factors — not technology — determine data-driven **decision-making** adoption

Brynjolfsson & McElheran (2016)

GPS data **reduces costs** when integrated into structured **analytical planning** workflows

Bø & Mjøsund (2022)

Literature review

Operational fuel Determinants

Winter +12%, autumn +10%, summer +4% relative to spring baseline conditions

Fontaras et al. (2008)

Speed, acceleration, gradient top predictors; trip-level modelling outperforms route averages

Chen et al. (2017)

Non-linear stop-spacing effect for suburban diesel buses; road gradient significant

Skrúcaný et al. (2025)

Stop-and-go driving adds up to +20% fuel vs free-flow — critical for suburban routes

Boriboonsomsin & Barth (2009)

ML Approaches to Fuel Prediction

XGBoost matches LSTM/NARX at R^2=0.99 — far lower computational overhead on ECU trip data

Canal et al. (2024)

MLR + SVM best accuracy-interpretability balance across 7 algorithms compared

Abdelaty & Mohamed (2021)

ANN with 7 aggregated trip-level inputs achieves R^2=0.966 without second-by-second CAN-bus data

Lorenc (2025)

LSTM best for instantaneous fuel — requires second-by-second telemetry

Grabowski et al. (2025)

Literature review

GAPS

Existing literature does not address these dimensions jointly

1. Linking predictive fuel analytics to fixed-tariff procurement
2. GLONASS telemetry integration in regional Russian bus operations
3. Pre-bid contract-level fuel cost forecasting for competitive tendering

H1 **District & Route Heterogeneity**

District-level operating conditions produce systematic, predictable differences in fuel consumption that a single normative coefficient cannot capture.

H2 **Non-Linear ML > Linear Methods**

Non-linear interactions between vehicle type, route conditions, and season make ensemble models significantly outperform linear regression.

H3 **Vehicle Identity as Independent Signal**

Individual vehicle identity (bus code) explains a material and independent share of trip-level fuel variance, comparable in magnitude to operating district

H4 **System Delivers Positive NPV**

Replacing the normative approach with a data-driven forecasting model yields a positive NPV over a 3-year horizon given Palmira's operational conditions.

LLC «Palmira» - Company Overview

Founded 2006, St. Petersburg & Leningrad Oblast

GLONASS installed

All buses, every 5–10 sec tracking

E1Card fuel cards

Corporate rates; transitioning to on-site TPZ

9 active licenses

ERA-GLONASS integrated

1.56 bn RUB revenue

+65.9% growth in 2025

0.41

Current liquidity

Strategic Context: PESTEL & SWOT

Main implication of PESTEL

Palmira already possesses the telemetry infrastructure required for predictive fuel analytics, but **rising fuel costs**, **regulatory pressure**, and **operational variability** make accurate forecasting strategically necessary.

Main implication of SWOT

Main competitive advantage - its existing operational data infrastructure, while its primary vulnerability is inefficient fuel-cost management under volatile market conditions.

AS-IS: Current Fuel Management Workflow

1
GLONASS Data Collection
GPS + FLS every 5–10 sec. Full telemetry recorded.

2
AM-23-r Norm
Fixed L/100km per class. No seasonal/district adjust.

3
Trip Executed
Actual vs planned — no real-time feedback or alerts.

4
Passive Reports
Downloaded on request only. No monitoring.

5
Tender Bid Prepared
Manual norm-based estimate

What Data is collected ✓

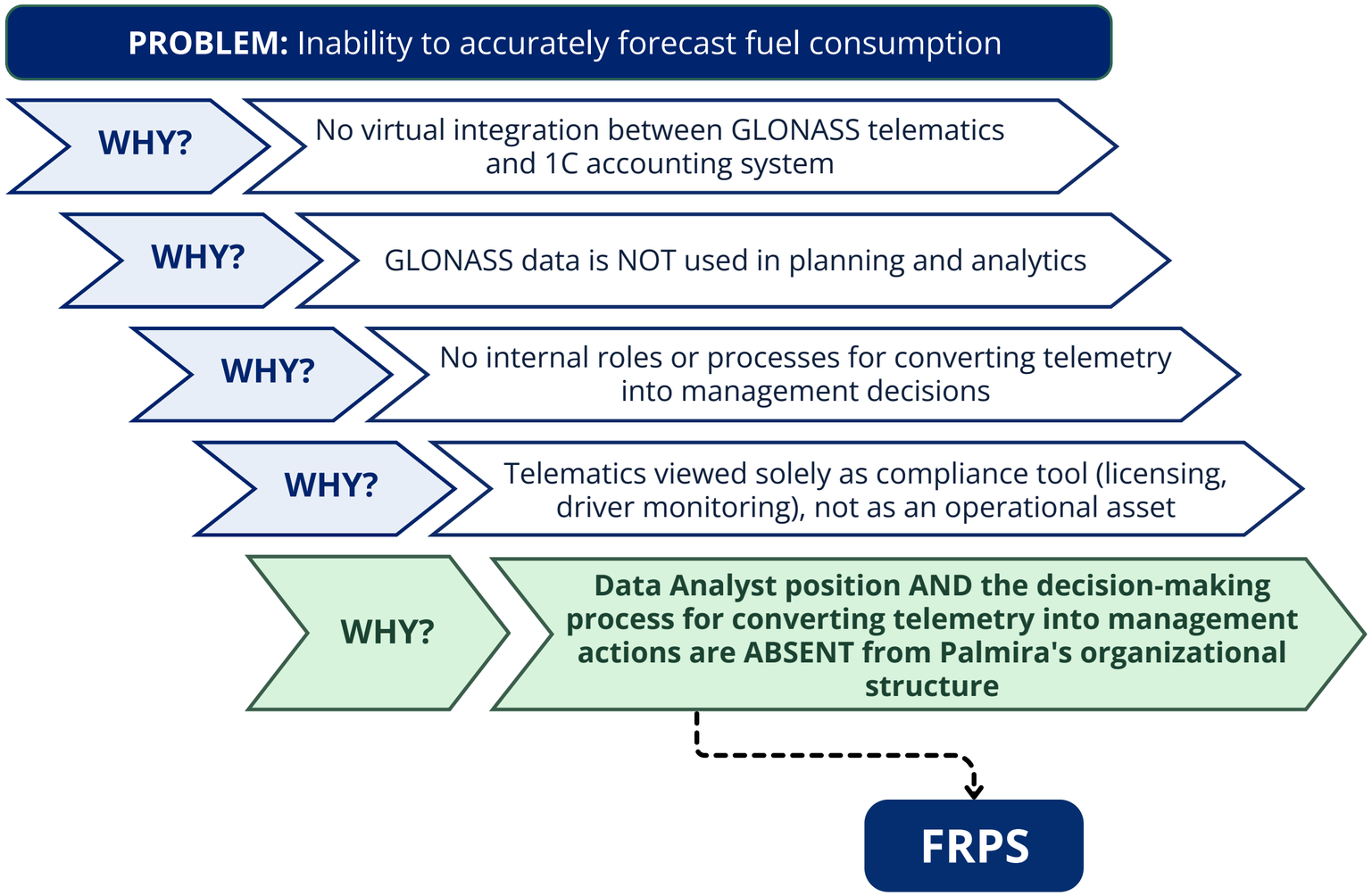
- Trip distance, start/end timestamps
- Actual fuel consumed (FLS sensor)
- Vehicle ID, route, GPS track
- Driver ID, tachograph records

What is missing ✗

- Integration with 1C accounting system
- Seasonal/district consumption benchmarks
- Pre-bid fuel cost forecasting tool

Analytical section

Root Cause Analysis: 5 WHY



Fuel Rate Prediction System (FRPS)

is not just an IT project
-
it is an **organisational capability development** initiative

The system is necessary but not sufficient; process redesign + role definition are equally critical.

Analytical section

Analytical Conclusion:

Palmira already collects large volumes of operational telemetry, but:

- data remains isolated,
- forecasting is absent,
- procurement planning relies on static norms.

the company requires a FRPS capable of converting telemetry into managerial decisions.

Dataset & Descriptive Statistics

70,132 trips	200 vehicles	7 districts	12 months	13 features

(Source of data: PivotGPS)

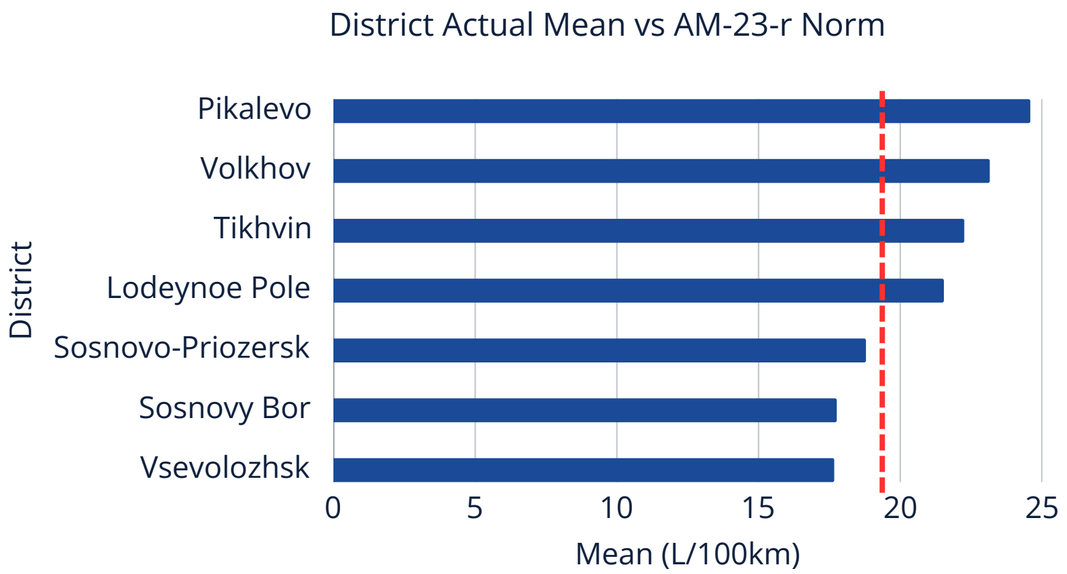
Variable	Mean	Median	Std Dev	Min	Max	CV (%)
Fuel consumed (L)	21.26	16.58	19.40	0.06	244.57	91.3
Distance (km)	109.45	83.14	97.55	0.50	697.05	89.1
Average (L/100km)	**20.14**	**18.66**	**8.39**	**5.00**	**45.38**	**41.6**
Consumption rate (L/100km)	19.11	20.00	7.66	10.00	40.00	40.1

Consumption rate = Normative rate (AM-23-r Order)

(Source: compiled by the authors on the basis of company's data)

Despite only 1.03 L/100km difference in means, the high standard deviations (8.39 and 7.66) reveal enormous trip-level heterogeneity that aggregate averages conceal

District Analysis



(Source: compiled by the authors on the basis of company's data)

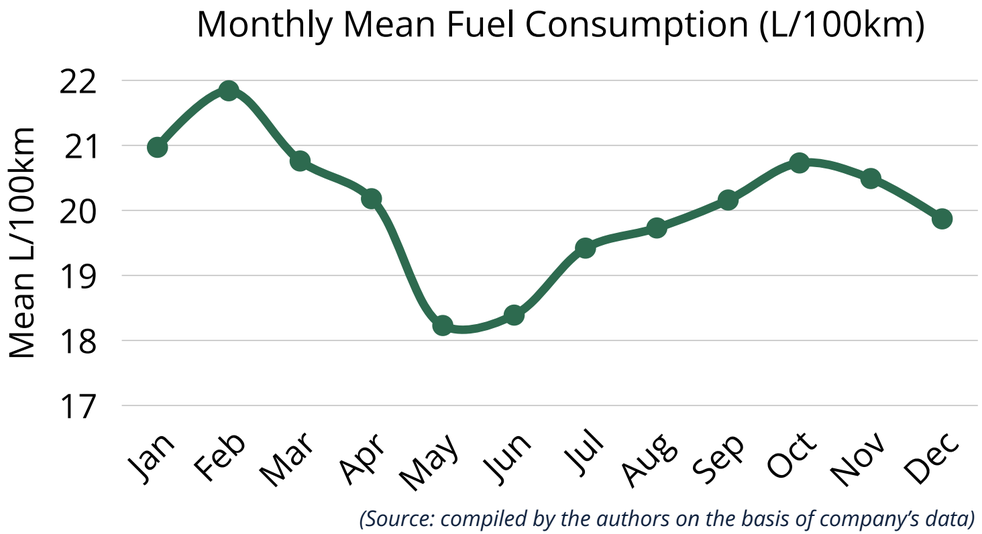
District	Trip Count	Mean (L/100km)	vs Norm	vs Fleet Mean
Pikalevo	1,951	24.59	**+5.48**	**+22.1%**
Volkhov	13,594	23.16	**+4.05**	+15.0%
Tikhvin	14,038	22.26	**+3.15**	+10.5%
Lodeynoe Pole	2,275	21.54	**+2.43**	+7.0%
Sosnovo-Priozersk	9,977	18.79	−0.32	−6.7%
Sosnovy Bor	10,975	17.76	**−1.35**	−11.8%
Vsevolozhsk	17,322	17.67	**−1.44**	**−12.3%**
Fleet total	**70,132**	**20.14**	**+1.03**	—

(Source: compiled by the authors on the basis of company's data)

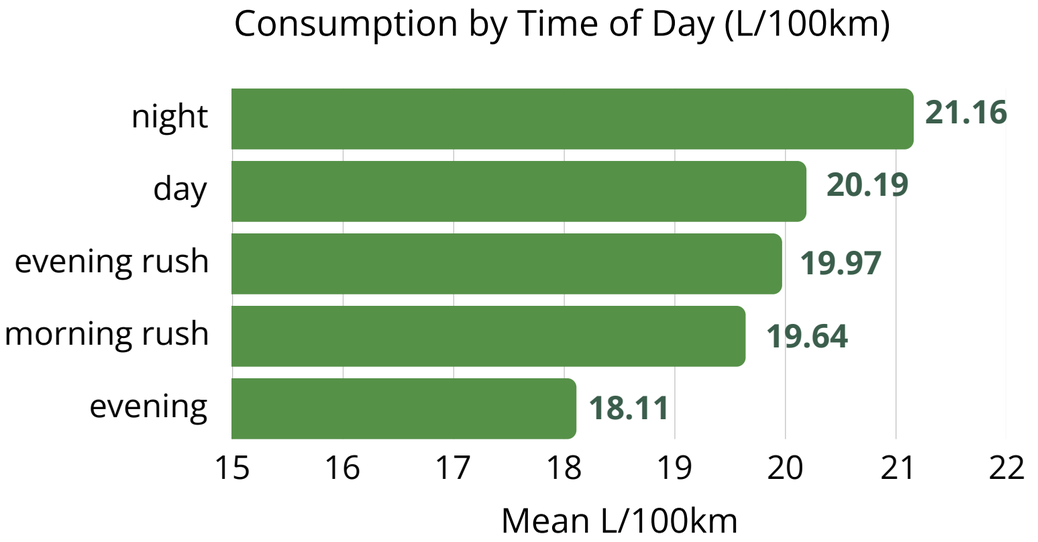
34.4% district spread | **6.92** L/100km range | **H1 ✓** confirmed

Three districts exceed the fleet mean by over 10% — any uniform normative rate will systematically underprice contracts in Tikhvin, Volkhov and Pikalevo.

Seasonal & Time-of-Day Effects



(Source: compiled by the authors on the basis of company's data)



(Source: compiled by the authors on the basis of company's data)

Winter consumption peaks 19.8% above the summer trough. A bid covering winter months without seasonal adjustment will underestimate fuel needs by approximately 1.5–2 L/100km.

Night operations consume the most fuel despite lighter traffic. A 3 L/100km spread across periods cannot be ignored in contract pricing.

Machine Learning Model Comparison

Model	MAE (L/100km)	MAPE (%)	R^2
Baseline — Normative Rate (AM-23-r)	8.00	46.30	-0.65
Lasso Regression	6.10	37.30	0.12
Linear Regression	6.10	37.30	0.12
Ridge Regression	6.10	37.30	0.12
SVM LinearSVR	6.00	34.50	0.08
ANN MLP	4.45	27.70	0.47
Decision Tree	4.35	26.90	0.47
Gradient Boosting	4.27	26.80	0.51
Random Forest	4.17	26.10	0.53
CatBoost	4.16	26.20	0.53
HistGradientBoosting	4.12	25.90	0.54
XGBoost (default)	4.07	25.50	0.54
LightGBM	4.02	25.30	0.55
Extra Trees	3.85	24.10	0.57
XGBoost GridSearch	**3.71**	**23.20**	**0.59**

(Source: compiled by the authors on the basis of company's data)

Selected Model

XGBoost GridSearch

GridSearchCV 5-fold, n_estimators ∈ {100,200}, max_depth ∈ {8,10}, lr ∈ {0.05,0.1}

53.6%

reduction in MAE vs AM-23-r norm
8.00 → 3.71 L/100km

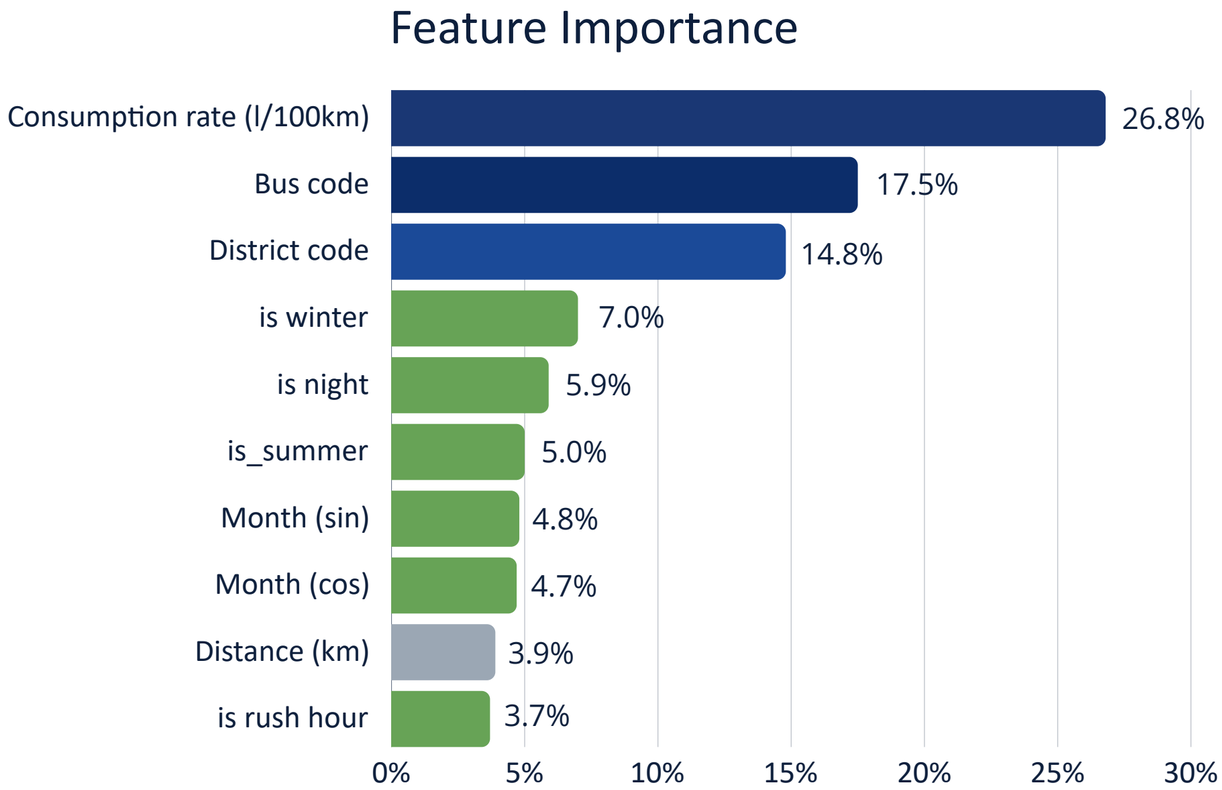
linear

H2 ✓

ensambles

XGBoost MAE 3.71 vs Linear 6.10 - 39% improvement; R^2 0.59 vs 0.12

Feature Importance — XGBoost GridSearch



(Source: compiled by the authors on the basis of company's data)

44.3%
Regulatory & Vehicle
Norm rate 26.8% + Bus code 17.5%

31.1 %
Seasonal & Temporal
winter + night + summer + month + rush hour

14.8%
District
District code as key route-environment signal

H3 ✓

Bus code importance 17.5% ≈ District code 14.8% — comparable independent signals

GLONASS Fuel Analytics - Strategic Transformation

Strategic Goal

Transform Palmira's GLONASS infrastructure from a passive compliance tool into an active fuel analytics instrument for competitive government contract bidding

1 Predictive Model

XGBoost trained on GLONASS telemetry; MAE ≤ 4.0 L/100km, R^2 ≥ 0.50 vs AM-23-r baseline

2 Contract Forecast Module

Trip-level predictions → total fuel volume & cost for any tender, with vehicle/monthly breakdown

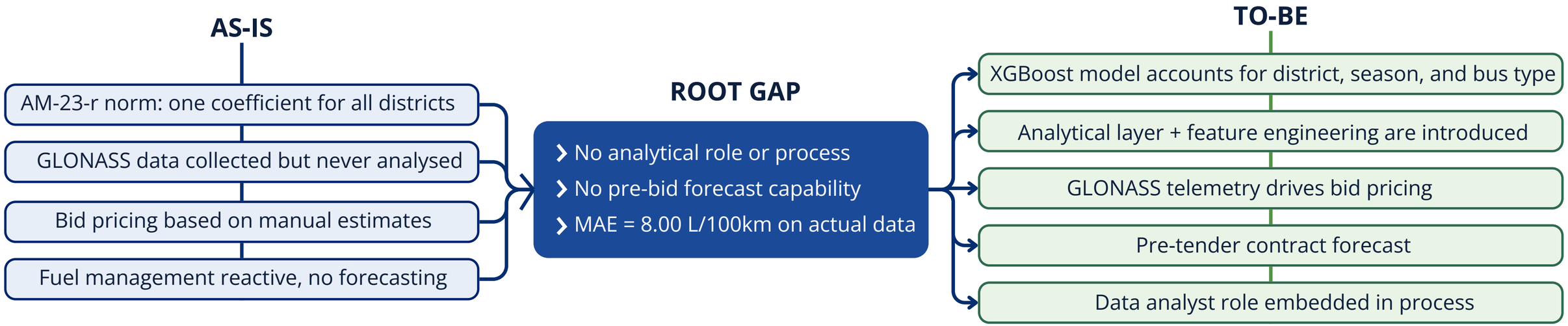
3 Reporting Output

Excel/CSV report with viability indicator in ≤10 min; support what-if scenario analysis

4 Documentation

BRD, feature engineering spec, hyperparameter log & retraining guide — maintenance by any analyst

AS-IS TO-BE FRAMEWORK



FRPS Architecture & Business Requirements

DATA LAYER
- PilotGPS CSV Export
- Vehicle Register
- District Register
- Contract Template

ANALYTICAL LAYER
- Feature Engineering
- XGBoost Training
- Model Inference
- Serialized Artifact

OUTPUT LAYER
- Fuel Volume (L)
- Fuel Cost (RUB)
- Viability Indicator
- Excel/CSV Report

USAGE LAYER
- Management Review
- What-If Scenarios
- Bid Price Input
- Procurement Plan

Business Requirements Document (BRD) — Key Blocks

Block 1: Data & Integration

BR-01A/B: CSV/API from PilotGPS
BR-02/03: Vehicle & district registers
BR-06/07: Validation & IQR cleaning

Block 2: Model

BR-09: Trip-level regression
BR-10: 13 features required
BR-13: Fixed seed, reproducibility

Block 3: Contract Forecasting

BR-15: Aggregate to contract total
BR-16: Viability indicator
BR-17: What-if scenario support

Block 4: Reporting

BR-18: Excel/CSV per-trip report
BR-19: MAPE uncertainty ranges
NFR-01: ≤10 min generation time

Evaluation of Architectural Alternatives

External Vendor (Opturan) ✗

- Covers driver/vehicle scheduling with AI optimisation the platform is oriented toward roster planning.
- The engagement did not result in an implemented solution.

Functional mismatch; no support for contract-level fuel forecasting; high customization costs.

Full 1C ERP Integration ✗

- Embed FRPS into Palmira's 1C system for automated data ingestion, reporting, and budget integration.
- Requires certified 1C developers and a dedicated project.

1C integration would require certified developers and complex coupling; FRPS is used for 3–5 tenders per year — unnecessarily complex and costly for such low frequency.

Real-Time API Dashboard ✗

- Live data ingestion via PilotGPS API, real-time forecasts on web dashboard.
- Duplicates PilotGPS monitoring, adds server and availability overhead.

- FRPS mismatch: output needed once per tender cycle, not continuously.
- Excessive complexity and limited business value for strategic bidding decisions.

Standalone Batch Python Solution ✓

- Manual CSV export from PilotGPS → XGBoost inference → Excel/CSV report.
- Runs on standard office hardware with no cloud dependencies.
- Versioned model artifact. Three inputs: historical telemetry, vehicle register, contract template.

Minimal infrastructure cost, runs on standard office hardware, maintained by one analyst — ideal for infrequent, high-stakes pre-bid decisions.

Contract Forecast Module Demonstration

3 buses , 15 Jun 2026 – 31 Dec 2027, 18.5 months, 26,555 trips, 245,210 km

Vehicle	FRPS L/100km	Norm L/100km	Δ vs Norm	FRPS Fuel (L)	FRPS Cost (RUB)
Т477УН198	17.03	16.0	+1.3	14,89	1,012,520
Т277УН198	18.08	16.0	+2.8	14,182	964,376
Т364УН198	23.0	16.0	**+7.0**	19,303	1,312,604
TOTAL	**20.6 avg**	**16.0**	**+4.6 avg**	**48,375**	**3,289,486**

(Source: compiled by the authors based on FRPS model output)

Normative Underestimate Risk:

Norm: 39,234 L = 2,667,912 RUB

FRPS: 48,375 L = 3,289,486 RUB

Gap: −9,141 L = −621,574 RUB (18.9%)

What-If Scenario Analysis

Fuel 68 RUB/L	→	3,289,486 RUB
Fuel 75 RUB/L (+10.3%)	→	3,628,125 RUB
Replace T364 with efficient bus	→	~2,900,000 RUB

Vehicle T364 alone accounts for 39.9% of contract fuel

Economic Justification — NPV & Scenario Analysis

Scenario Analysis				
Scenario	**Benefit/yr**	**Payback**	**NPV 3yr**	**ROI 3yr, %**
Conservative (30%)	6.72M	81 days	11.8M	720
Base (50%)	11.2M	44 days	21.6M	1310
Optimistic (70%)	15.7M	30 days	31.3M	1900
Fuel price +15%	12.9M	38 days	25.3M	1530
Fuel price −15%	9.5M	53 days	17.9M	1080

Additional benefits:

- Reputational risk avoidance
- Data foundation for future tariff negotiations with government customers
- Enables proactive procurement timing (seasonal patterns)

Metric	Value
WACC	18%
Implementation Cost	1.2M RUB
Maintenance Cost	0.15M RUB/yr
MAE Improvement	4.29 L/100 km
Mileage	7,675,700 km/yr

(Source: compiled by the authors)

H4 ✓

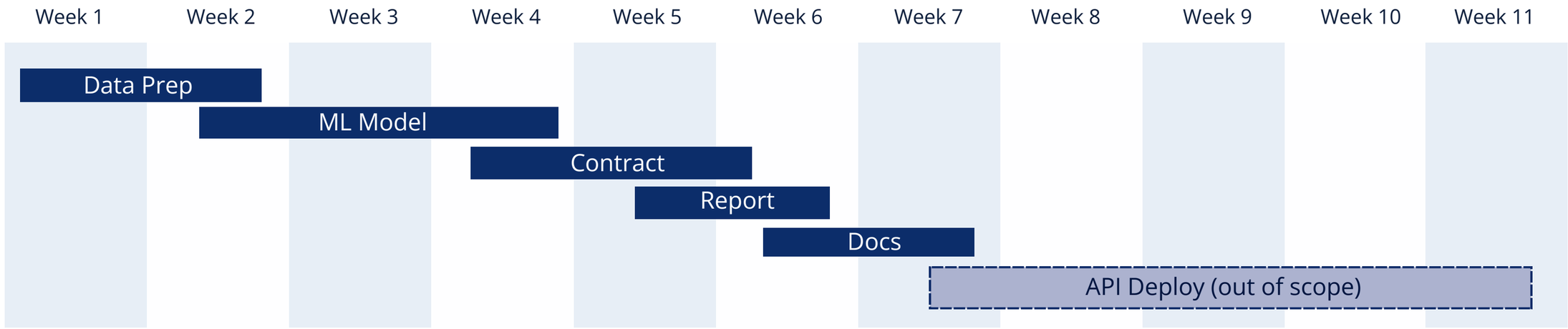
Positive NPV in ALL scenarios

KPI Framework

KPI	Baseline	Target	Measurement
Model MAE (L/100km)	8.00 (AM-23-r)	**≤ 4.00 ✓ Achieved**	Test-set eval each retrain
Model R^2	−0.65 (AM-23-r)	**≥ 0.50 ✓ Achieved**	Test-set eval
Contract cost estimation error	±15–25%	±10% vs actual	Post-contract reconciliation
Report generation time	N/A	≤ 10 minutes	User acceptance testing
FRPS adoption rate	0%	≥ 80% of bids	Project Manager tracking log
Prevented losses / year	0 RUB	≥ 400,000 RUB	Post-bid retrospective

(Source: compiled by the authors)

Schedule of Control Events

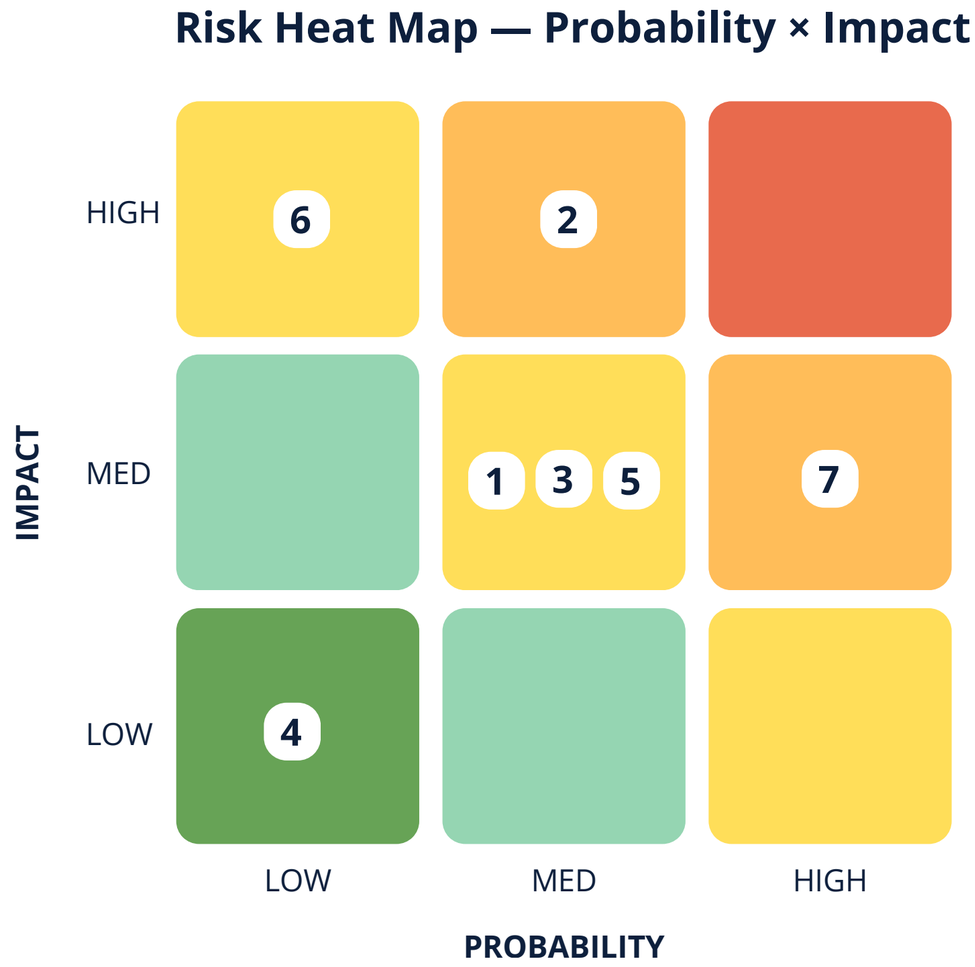


(Source: compiled by the authors)

Project Risks

#	Risk	Prob	Impact	Mitigation
1	GLONASS CSV Errors	Med	Med	BR-06 validation
2	Management Trust in ML	Med	**High**	Per-trip audit trail
3	Input Template Errors	Med	Med	Dropdown validation
4	AM-23-r Regulatory Change	**Low**	**Low**	Configurable parameter
5	Model Distributional Shift	Med	Med	Annual retraining
6	Key Staff Departure	**Low**	**High**	Full documentation
7	Fuel Price Volatility	**High**	Med	Scope: volume only

(Source: compiled by the authors)



Conclusion

C1

Static norms structurally fail

AM-23-r: MAE = 8.00 L/100km. District spread of 34.4% is systematic, not incidental — H1 confirmed.

C2

XGBoost reduces error by 53.6%

MAE 3.71 vs 8.00. Non-linear ML essential for heterogeneous regional fleets — H2 & H3 confirmed.

C3

NPV positive across all scenarios

3-yr NPV: 11.8–31.3M RUB. Payback ≤ 82 days. Contract demo shows 621K RUB undershoot risk — H4 confirmed.

C4

Barrier is organisational, not technical

Data already exists. Missing: analyst role + process to convert GLONASS telemetry into bid inputs.

53.6%
MAE reduction

≤ 82
days payback

31.3M
RUB NPV (opt.)

0.59
R^2 achieved

Conclusion

Current Limitations

$R^2 = 0.59$ — 41% unexplained

No driver behaviour variables. No vehicle maintenance state. Trip-aggregated data limits granularity.

Manual data ingestion

BR-01B (API integration) deferred to Phase 5. Current manual CSV export is sufficient for 3–5 tenders/year use case.

Scope: volume uncertainty only

Total fuel cost = volume × price. FRPS forecasts volume; price component remains exogenous. Forward fuel card agreements recommended separately.

200 vehicles / 7 districts boundary

Model cannot extrapolate to new routes or new vehicle registrations without retraining. Annual retraining recommended.

Future Research Directions

Direction 1: CAN-bus telemetry

Activating sub-trip CAN-bus exports would allow driver behaviour features, potentially explaining a portion of the remaining 41% variance. (Ping et al., 2019)

Direction 2: PilotGPS API (BR-01B)

Automated API-based data ingestion enables continuous model retraining and adaptation to fleet changes without manual intervention.

Direction 3: Multi-carrier study

Cross-carrier validation across Leningrad Oblast operators would enable evidence-based tariff benchmarking under 44-FZ — filling the identified research gap.

Summary

The Problem 1

- Diesel prices are constantly rising
- Palmira uses a single Ministry of Transport coefficient to price government tenders
- Under 44-FZ fixed-tariff contracts, two costly outocmes.

Both underbidding and overbidding destroy value - and the current tool makes both inevitable.

Research Question 2

What system requirements, data inputs and analytical models are needed to build a fuel forecasting system for Palmira that estimates fuel costs before government contract bidding?

The literature covers telematics, fuel determinants and ML separately — but no study connects them to fixed-tariff procurement. This is the gap we fill

Analytical Section 3

70,132 clean trip records across 7 districts revealed a 34.4% spread in district consumption, a 19.8% seasonal swing, and systematic time-of-day differences. The normative rate sits near the fleet mean but fails on individual contracts.

H1 confirmed. One coefficient cannot represent 7 districts × 4 seasons × 200 vehicles.

Modelling Results 4

15 models compared. XGBoost GridSearch selected: MAE 3.71, R^2 0.59 — a 53.6% improvement over the normative baseline. The top three features (norm rate, bus code, district code) explain 59% of predictive signal. Vehicle identity and district were near-equal in importance.

H2 ✓ non-linear beats linear, H3 ✓ each bus has a measurable consumption fingerprint

Project Section 5

FRPS is a standalone batch Python pipeline: PilotGPS CSV → XGBoost inference → Excel/CSV report in ≤10 minutes. Specified through a BRD (19 functional + 6 non-functional requirements). Contract №1 demo: the normative approach would have missed 9,141 litres — 621,574 RUB of unaccounted cost.

Three vendor alternatives evaluated and rejected before selecting the standalone batch architecture

Conclusions 6

NPV 11.8–31.3M RUB across all scenarios, payback ≤ 82 days. The core finding: the barrier was never the data or the technology — it was the absence of a process to convert telemetry into decisions. FRPS creates that process.

H1 H2 H3 H4 all confirmed, Barrier is organisational, not technical

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Saint-Petersburg school of Economics and Management

International Program in Business and Economics

Data-Driven Company

Developing the forecast system of fuel consumption for the bus fleet considering routing network

Presented by: Sergeeva Viktoria, Ivanova Mariia, Terekhova Anastasia

Supervizor: Strimovskaya Anna Viktorovna
Seniour lecturer, Ph.D.

St. Petersburg, 2026

Appendix

PESTEL ANALYSIS

Factors	Influence on the company
Political	1. Palmira is obligated to comply with licensing requirements, transportation safety, and the technical condition of its rolling stock. 2. Participation in government tenders and contracts (under Federal Law No. 44-FZ) allows the company to secure stable orders from municipal and corporate clients, which forms the basis of its revenue. 3. Sanctions and import restrictions create risks, as many of the company's buses and spare parts are manufactured abroad, which can lead to higher prices and delivery delays. 4. Regional government transport policies also directly impact business profitability, as the company works closely with the Leningrad Oblast Transport Committee.
Economic	1. For the transportation company, fuel is one of the largest operating expenses. Rising diesel prices significantly increase operating costs within the company. 2. Inflation and rising transport costs lead to higher transportation costs. However, as the economy grows, so does the demand for corporate employee transportation. 3. Intercity and charter passenger transportation is affected by income levels; for example, during times of crisis, passengers may forgo comfortable but expensive routes.
Social	1. Increased social mobility and pronounced weekday seasonality due to commuting, as well as year-round seasonality due to the summer, are increasing demand for passenger transportation, providing opportunities for network expansion. 2. Demographic changes, such as the population concentration in St. Petersburg, may lead to a gradual decline in demand on outlying routes in the Leningrad Oblast, necessitating route adjustments. 3. Reputation and customer reviews directly influence the attraction of new customers, as a company's positive image is an important intangible asset, especially in the charter and children's transportation segments.

Appendix

PESTEL ANALYSIS

Technological	1. The development of GPS and monitoring systems such as GLONASS and tachographs allows the company to monitor routes, fuel consumption, and service quality. 2. The Avtokonduktor system, which provides transparent analytics and real-time reporting, is used for data management. Each bus is equipped with modern equipment, including stereoscopic passenger counting cameras and cashless payment validators. 3. Updating the fleet with modern buses reduces maintenance costs and improves safety, but requires significant investment.
Environmental	1. Seasonal temperature fluctuations in the Leningrad Oblast (from -24°C to +31°C) have a well-documented impact on bus fuel consumption. In winter, fuel consumption increases by 12%, which leads to an increase in CO2 emissions. Stricter environmental regulations require the company to renew its fleet and reduce emissions, creating additional investment burdens. 2. Since the company operates a significant fleet of diesel-powered vehicles, its activities are regulated by Federal Law No. 96-FZ. Compliance with these environmental standards is a mandatory requirement for Palmira to obtain a license. 3. Environmental inspections, for example, by Rosprirodnadzor and the Committee for Natural Resources carry the risk of fines and restrictions on operations for non-compliance.
Legal	1. The company's tax liabilities are determined primarily by the transport tax, calculated in accordance with Leningrad Oblast Law No. 51-OZ. 2. Since the company accepts cash payments from passengers for tickets directly on board the bus, it is required to use cash registers and transmit fiscal data to the Federal Tax Service (Federal Law No. 54). 3. Labor legislation obligates the company to strictly adhere to driver work and rest regulations. Violations are punishable by large fines and may lead to license suspension. Legislation on passenger transportation (Federal Law No. 259, the Charter of Automobile Transport) governs transportation contracts, liability for the life and health of passengers, and baggage regulations. 4. Transportation safety requirements include regular technical inspections, personnel qualification checks, and mandatory video surveillance. 5. Tax regulations such as changes in transport tax rates, benefits for carriers, and depreciation deductions impact the company's financial performance. 6. The company is required to comply with Federal Law No. 152 on personal data, as its buses are equipped with a video recording system for the driver's cabin and passenger compartment.

Appendix

SWOT ANALYSIS

Strengths	Weaknesses
1. Passengers can monitor buses equipped with GLONASS/GPS systems, tachographs, and video surveillance cameras in real time, ensuring continuous collection of primary data on the driver's movements, location, and performance. 2. The company's fleet consists of over 300 vehicles (Yutong, NEFAZ, PAZ, Gazelle CITY, and Higer), allowing for flexible response to various types of orders. 3. The long-term relationship with the Leningrad Oblast Transport Committee and regular participation in tenders in accordance with Federal Law No. 44-FZ ensure predictable income, creating a financial basis for investing in operational analytics (Russian Federation. (2013). Federal Law No. 44-FZ of April 5, 2013 “On the Contract System in the Sphere of Procurement of Goods, Works, Services for State and Municipal Needs”. Kontur.Normativ. Available from: [https://normativ.kontur.ru/document?moduleId=1&documentId=503423&ysclid=mpj1zw8nt0492462539]) . 4. The company holds 9 valid licenses and is integrated with the ERA-GLONASS government platform, mitigating regulatory risks. 5. Existing access to discounted corporate fuel pricing via E1Card, reducing exposure to retail fuel prices.	1. The company's operations are highly dependent on the prices of fuel, spare parts, and vehicle maintenance. All of these costs are subject to market fluctuations, inflation, and sanctions. The diverse fleet complicates the standardization of standards and repairs. 2. Its geographic presence is limited to St. Petersburg and the Leningrad Oblast, limiting growth opportunities and making the business sensitive to regional politics. 3. The company faces reputational risks. Any flight delays, traffic accidents, or passenger complaints are instantly disseminated via online maps and social media, directly impacting customer trust and potentially leading to lost orders, especially in the charter and children's transportation segments, where safety is a priority. 4. There is intense competition from federal carriers and aggregators, and a significant portion of the company's revenue is generated through government tenders, so the loss of one major contract could significantly impact its financial performance.
Opportunities	**Threats**
1. Growing domestic tourism and demand for corporate transportation allow for the expansion of charter and excursion services. 2. Participation in state and municipal programs provides access to subsidies for fleet renewal, particularly for eco-friendly vehicles, which will reduce fuel costs in the long term. 3. Expanding the route network to neighboring regions and expanding into the B2B segment, for example, employee transportation and school routes, is also possible, providing stable, long-term contracts. 4. Ongoing transition from outsourced refuelling (E1Card network) to on-site modular fuel stations (TPZ), expected to significantly reduce fuel procurement costs.	1. There is a diesel fuel price volatility. A 5-ruble increase per liter results in an additional 2.5-3.5 million rubles in annual costs based on current mileage. 2. Tighter government regulations and environmental requirements will require costly fleet upgrades. 3. The shortage of qualified drivers in the labor market is exacerbated by an aging workforce and the profession's low appeal to young people. Additionally, competition from car-sharing, taxis, and rail transport is increasing, especially on short-haul and high-speed routes. 4. Road quality problems in the Leningrad Oblast increase fuel consumption and bus wear. 5. Dependence on imported spare parts for Chinese and European buses under sanctions carries the risk of downtime and increased repair costs. There are also risks of reputational damage due to schedule delays or a decline in service quality.

Appendix

Full descriptive statistics for all quantitative variables

	mean	std	min	25%	50%	75%	max	skew	kurtosis	IQR	CV%
Fuel consumption (l)	21.26	19.40	0.06	6.99	16.58	28.22	244.57	1.95	6.00	21.23	91
Distance (km)	109.45	97.55	1.00	41.78	83.14	143.56	697.05	1.87	4.17	101.78	89
Average (l/100 km)	20.14	8.39	5.00	14.37	18.66	23.83	45.38	0.89	0.52	9.46	42
Consumption rate (l/100km)	19.11	7.66	10.00	14.00	20.00	20.00	40.00	0.91	0.52	6.00	40
Hour	10.57	5.48	0.00	6.00	10.00	15.00	23.00	0.30	-0.85	9.00	52
Day of the week	2.89	1.97	0.00	1.00	3.00	5.00	6.00	0.06	-1.21	4.00	68
Month	6.19	3.47	1.00	3.00	6.00	9.00	12.00	0.17	-1.24	6.00	56
Month (sin)	0.09	0.73	-1.00	-0.50	0.00	0.87	1.00	-0.17	-1.53	1.37	827
Month (cos)	0.01	0.68	-1.00	-0.50	0.00	0.50	1.00	0.02	-1.42	1.00	11250
Weekend	0.26	0.44	0.00	0.00	0.00	1.00	1.00	1.12	-0.74	1.00	171
is winter	0.25	0.44	0.00	0.00	0.00	1.00	1.00	1.13	-0.73	1.00	171
is summer	0.22	0.41	0.00	0.00	0.00	0.00	1.00	1.36	-0.16	0.00	189
is night	0.25	0.44	0.00	0.00	0.00	1.00	1.00	1.13	-0.72	1.00	172
is rush hour	0.27	0.44	0.00	0.00	0.00	1.00	1.00	1.05	-0.90	1.00	165

Appendix

Month statistics

Month	Mean	Median
1	20.97	19.88
2	21.84	20.84
3	20.76	19.11
4	20.18	18.80
5	18.23	16.67
6	18.39	16.07
7	19.42	17.52
8	19.73	18.19
9	20.16	18.63
10	20.73	19.44
11	20.49	19.11
12	19.87	18.71

Appendix

Data Cleaning

174,913
Raw trip records

→

99,990
After deduplication

Step	Criteria	Threshold / Condition	Rows Removed
1	Missing data	No district or no class info	6,880 + 1,337
2	Physical constraints	Speed > 100 km/h Distance < 1 km	3,007 3,841
3	IQR outlier (target variable)	Below Q1 – 1.5·IQR Above Q3 + 1.5·IQR	6,876
4	Rare vehicle	< 10 recorded trips	2

Appendix

Data Cleaning

99,990
After deduplication

↓

70,132
trips

Cleaning Step	Records Removed	Reason
No district information	6,88	Missing geographic reference data
No class information	1,337	Missing vehicle classification data
Speed > 100 km/h	3,007	Physically implausible, likely GPS errors
Distance < 1 km	3,841	Depot movements, not revenue service
Duration > 16 hours	4,358	Recording errors or multi-day aggregation
Average (l/100 km) < 5 L/100 km	3,557	Recording errors, not real bus fuel consumption
IQR outliers (L/100km > 46.05)	6.876	Statistical outliers, sensor anomalies
Vehicles with < 10 trips	2	Insufficient history for estimation
Total removed	**19,236 (19.2%)**	
Final clean sample	**70,132**	**Used for all subsequent analysis**

Appendix

Project Objectives, KPIs, and Target Values

Objective	KPI	Target Value
Develop and validate a trip-level fuel consumption prediction model trained on Palmira's historical GLONASS telemetry	Model MAE (L/100km); Model R^2	MAE ≤ 4 L/100km; R^2 ≥ 0.50 (≥ 50% reduction vs. AM-23-r normative baseline)
Build a contract-level aggregation module that converts trip predictions into total fuel volume and cost for any prospective tender specification	Forecast completeness; Breakdown granularity	100% of specified trips predicted; output by vehicle and calendar month
Produce a structured Excel/CSV report providing management with fuel cost exposure estimates for each prospective contract	Report generation time; Format compliance	≤ 10 minutes; report contains all mandatory fields per BR-18
Deliver complete technical documentation enabling future system maintenance independent of the original development team	Documentation completeness	BRD, feature engineering specification, model hyperparameters, retraining guide, and user manual delivered.

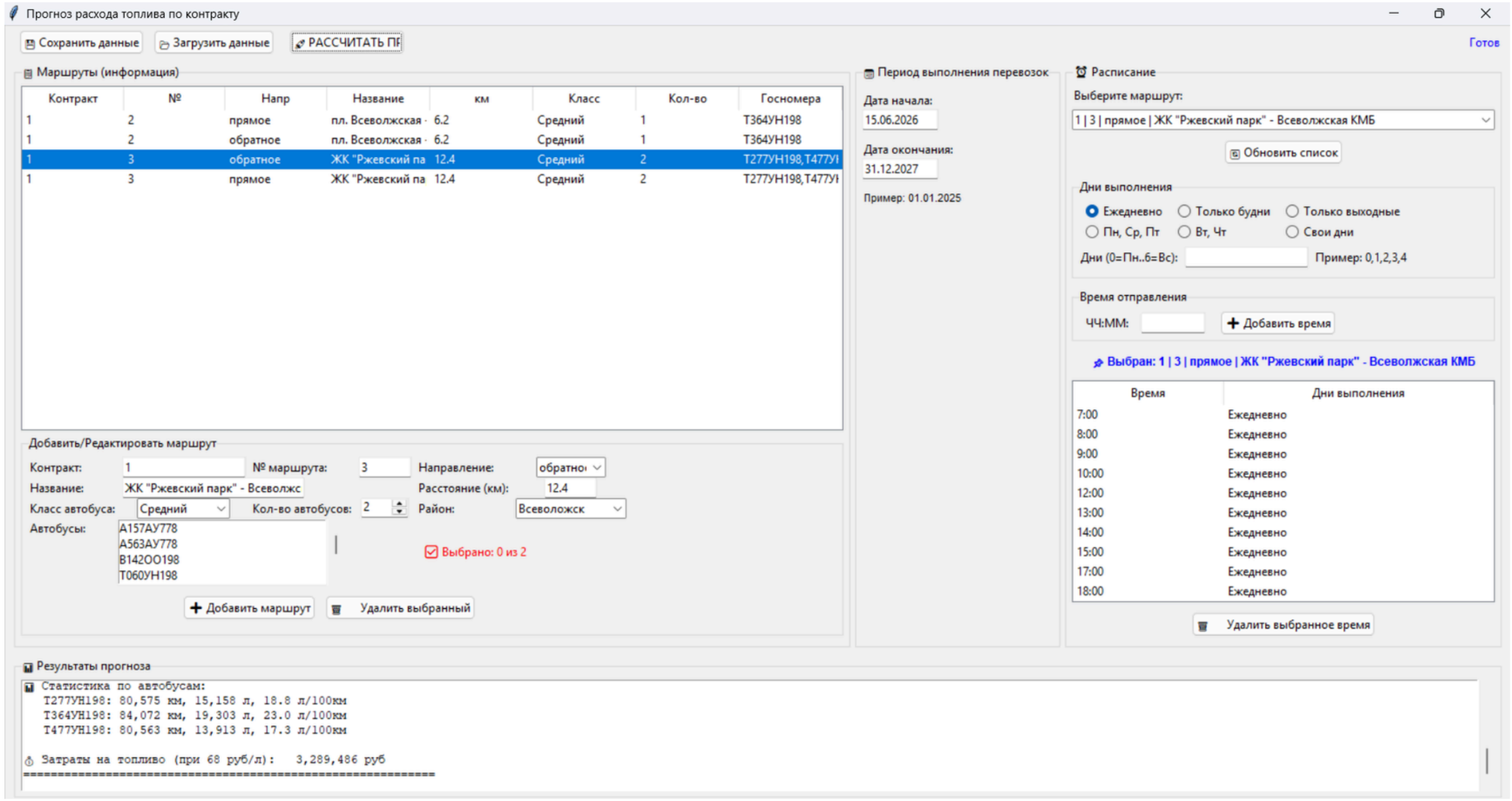
Appendix

Correlation of Numerical Variables



Appendix

Fuel Rate Prediction System (FRPS): Contract Forecasting Dashboard



Appendix

Economic Scenario Analysis calculations

Appendix

Project Risks

Risk	Likelihood	Impact	Owner	Mitigation
GLONASS CSV export contains structural errors or missing columns	Medium	Medium	**Data & ML Analyst**	Implement input validation (BR-06) and outlier removal (BR-07); maintain schema documentation
Management does not trust ML predictions over familiar normative rates	Medium	High	**Project Manager**	Provide full per-trip breakdown in report for auditability; run parallel comparison of FRPS vs normative on historical actuals during validation phase
Contract input errors in manual Excel template	Medium	Medium	**Business & Reporting Analyst**	Template validation on load (BR-04); clear error messages; pre-populated dropdown lists for district codes and bus types
Regulatory change to AM-23-r normative rates	Low	Low	**Business & Reporting Analyst**	Norms stored as a configurable parameter in the vehicle register (BR-02); update requires only data entry, not code change
Model accuracy degrades on future data (distributional shift)	Medium	Medium	**Data & ML Analyst**	Annual model retraining protocol; performance monitoring dashboard comparing predicted vs actual on completed contracts
Key technical personnel leave before documentation is complete	Low	High	**Project Manager**	Full technical documentation and user guide; handover package includes model file, feature engineering script, and step-by-step retraining guide
Fuel price volatility undermines cost forecast validity	High (external)	Medium	**Financial Director**	Clearly communicate scope limitation: FRPS provides volume forecast, not price forecast; recommend forward fuel card agreements separately

Appendix

Roles and Key responsibilities

Role	**Assigned to**	Key responsibilities
Project Manager	**Ivanova Mariia**	Coordination, communication, documentation, risk register, success criteria
Data & ML Analyst	**Terekhova Anastasia**	Data preparation, feature engineering, model training and evaluation
Business & Reporting Analyst	**Sergeeva Viktoria**	Contract logic, viability assessment, Excel/CSV report generation, what-if scenarios

Appendix Business Requirements Document (BRD)

Block 1. Data Collection and Integration

ID	Requirement	Priority	Rationale
BR-01A	The system MUST ingest historical telemetry data from GLONASS via manual CSV export from PilotGPS, covering a minimum of 12 months of trip records.	Must	GLONASS data exist but are not used analytically. For this term paper, manual export is used.
BR-01B	The system SHOULD implement a programmatic API integration with PilotGPS for automated data ingestion. When implemented, the system SHALL fetch new data at scheduled intervals (e.g., every 6 months) and trigger model retraining.	Should	Production target. API exists per vendor documentation; implementation deferred due to project timeline
BR-02	The system MUST maintain a vehicle register with: vehicle ID, bus type (Bus code), AM-23-r baseline norm (l/100 km).	Must	Required for feature engineering.
BR-03	The system MUST maintain a route district register with: district code (District).	Must	Categorical feature for spatial heterogeneity.
BR-04	The system MUST accept contract specifications (planned trips, bus assignment, season of execution) via a predefined Excel/CSV template. The template SHALL include mandatory columns (e.g., trip count or schedule, route district, bus type, season flag).	Must	Low tender frequency (3–5 per year) and high variability in contract formats make full automation economically infeasible. Manual entry via template ensures structured input.
BR-05	The system MAY support semi-automatic extraction from structured DOCX tables (e.g., route schedule tables provided in the tender document). Extracted values SHALL be presented to the user for review and confirmation before being used in forecasting.	Could	Reduces manual entry effort when the tender document contains well-formed tables. User review ensures correctness; incomplete or malformed tables fall back to manual entry.

Appendix Business Requirements Document (BRD)

Block 1. Data Collection and Integration

BR-06	The system MUST validate any input data file (both historical telemetry CSVs for training and contract specification templates for forecasting) for basic correctness: required columns present, correct data types, non-negative values where applicable (e.g., distance, trip count), and valid categorical values (e.g., district codes, bus types, season flags). If validation fails, the system SHALL return a clear error message identifying the issue and SHALL NOT proceed with training or forecasting until the error is resolved.	Must	Prevents errors propagating from incorrect input data into model training or business forecasts.
BR-07	The system MUST remove statistical outliers from the training data before model training. Outliers SHALL be defined as trips with fuel consumption per kilometer falling outside the interquartile range (IQR) method, or outside configurable domain-specific thresholds.	Must	Prevents extreme values (measurement errors, abnormal events) from distorting model training.
BR-08	The system SHOULD preserve the trained model artifact from each training run, together with its version identifier (timestamp or sequential number) and performance metrics (MAE, MAPE, R^2 on holdout set). A previous version SHALL be restorable for comparison or rollback if a newer model degrades in performance	Should	Enables model versioning and rollback capability when retraining is automated (BR-01B). This is a recommended future enhancement.

Appendix Business Requirements Document (BRD)

Block 2. Predictive Model

ID	Requirement	Priority	Rationale
BR-09	The system MUST predict fuel consumption (liters) for a single trip before the trip occurs, using a regression model trained on historical GLONASS telemetry data. The final model SHALL be selected as the best-performing candidate based on validation metrics.	Must	Core ML deliverable. Model selection is based on empirical performance comparison.
BR-10	The system MUST use the following feature set for prediction: trip distance, hour of day, day of week, weekend flag, month (sin/cos encoded), winter flag, summer flag, night flag, rush hour flag, bus code (categorical), route district code (categorical), and AM-23-r normative fuel rate as a reference feature.	Must	Feature set derived from available GLONASS data.
BR-11	The system MUST compare at least two regression models. The comparison SHALL be performed using MAE, MAPE, and R^2 on an identical test split.	Must	Establishes performance improvement over a simple baseline.
BR-12	The system SHALL select the final production model as the candidate with the lowest validation error (e.g., lowest MAE). The selected model and its performance metrics SHALL be documented.	Must	Ensures systematic, reproducible model selection.
BR-13	The system MUST use a reproducible train-test split (random seed fixed). The split method (e.g., time-based or random) SHALL be documented.	Must	Prevents data leakage. Fixed random seed ensures reproducibility (academic standard)
BR-14	The system MUST NOT use real-time telemetry, tachograph data, or fuel card reconciliation for trip-level prediction. Prediction is based solely on historical data and contract inputs.	Will Not	FRPS supports bidding decisions, not operational monitoring. Real-time and post-hoc data are out of scope.

Appendix Business Requirements Document (BRD)

Block 3. Contract Forecasting and Viability Assessment

ID	Requirement	Priority	Rationale
BR-15	The system MUST aggregate trip-level fuel consumption predictions to estimate total fuel consumption (liters) for a prospective government contract based on the contract specification (routes, trip counts, bus types, season).	Must	Directly answers the research question: total fuel cost per contract.
BR-16	The system MUST calculate total fuel cost (liters × configurable fuel price), compare it against the government fixed tariff, and output a viability indicator ("Profitable", "Marginal", or "Unprofitable") based on configurable thresholds.	Must	Converts ML prediction into business decision support. Viability indicator simplifies management review.
BR-17	The system SHOULD support what-if scenario analysis: management can modify input parameters (bus type, season, fuel price, tariff structure) and see updated viability outcome.	Should	Strategic decision support. Enables sensitivity analysis before bidding.

Appendix Business Requirements Document (BRD)

Block 4. Reporting and Output

ID	Requirement	Priority	Rationale
BR-18	The system MUST generate an Excel/CSV report containing: (a) per-trip predicted fuel consumption (liters), (b) total predicted fuel consumption for the contract (liters), (c) total predicted fuel cost, (d) comparison with government tariff, (e) viability indicator (Profitable / Marginal / Unprofitable).	Must	Core output format. Management requires structured, auditable data for bidding decisions.
BR-19	The system SHOULD include prediction uncertainty estimates (e.g., confidence intervals or MAPE range) for total contract fuel consumption.	Should	Supports risk-aware bidding. Management can see best-case / worst-case scenarios.

Appendix

Non-Functional Requirements

ID	Requirement	Priority	Rationale
NFR-01	The system MUST produce the forecast report within 10 minutes of receiving the contract specification input.	Must	Batch processing is sufficient for bidding (tenders are prepared over days, not minutes). 10 minutes allows for model inference and report generation.
NFR-02	The system MUST ensure reproducibility: the same input data and contract specification SHALL produce identical output on repeated runs (deterministic model training with fixed random seed).	Must	Auditability for bidding. Management must trust that results are consistent and verifiable.
NFR-03	The system MUST NOT require 24/7 availability or real-time data ingestion. Offline batch operation is sufficient.	Will Not	FRPS is used for tender preparation, not daily fleet operations.
NFR-04	The system SHOULD run on standard office hardware (8GB RAM, any OS) without cloud dependencies.	Should	Palmira may not have dedicated ML infrastructure. Low barrier to adoption.
NFR-05	Historical telemetry data SHALL be stored locally on the analyst's machine (e.g., project folder). No external cloud storage is required.	Must	Simplicity. No additional compliance burden (FL No. 152-FZ is not triggered because no personal driver data is stored).
NFR-06	The system SHOULD be scripted in Python with documented dependencies (requirements.txt or environment.yaml) to ensure portability to another analyst's machine.	Should	Enables knowledge transfer, future maintenance, and potential handover to Palmira's IT team.

Appendix

пальмира
транспортная компания

Общество с ограниченной ответственностью «Пальмира»
Юридический адрес: 195299, г. СПб, Руставели, 75, лит А
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ИНН 7806345155 КПП 780401001
ОКПО 9676463 ОГРН 5067847374293

Отзыв
от представителя ООО «Пальмира» о работе проектной команды, состоящей из студентов группы БМББЭ2307С

Ивановой Марии Дмитриевны
Сергеевой Виктории Викторовны
Тереховой Анастасии Васильевны

Команда продемонстрировала высокий уровень самостоятельности и инициативы на всех этапах разработки системы прогнозирования расхода топлива для автобусного парка. Студенты самостоятельно сформулировали цели и задачи проекта и тщательно отобрали инструменты анализа. В течение всего проекта проводились регулярные встречи с представителями компании с использованием существующих телематических платформ.

График выполнения курсового проекта был полностью соблюден. Все этапы, от сбора необходимых данных до расчета экономической эффективности, были завершены в утвержденные сроки. Структура и содержание отчета полностью соответствуют требованиям к курсовой работе в данной области. Форматирование отчета также было безупречным.

Вывод: Курсовой проект проектной команды полностью соответствует требованиям компании «Пальмира». Компания высоко ценит качество выполненного исследования и рекомендует разработанные требования к системе прогнозирования расхода топлива для пилотного внедрения на маршрутах Ленинградской области.

26.05.2026г.

Генеральный директор ООО «Пальмира» ***М.А. Жукова***



пальмира
транспортная компания

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ЗАДАНИЕ НА ВЫПОЛНЕНИЕ ПРОЕКТА ДЛЯ ООО «Пальмира»
В РАМКАХ НАПИСАНИЯ КУРСОВОЙ РАБОТЫ

Следующие студенты группы БМББЭ2307С, Иванова Мария Дмитриевна, Сергеева Виктория Викторовна и Терехова Анастасия Васильевна, участвуют в реализации следующего проекта «Developing the forecast system of fuel consumption for the bus fleet considering routing network» по запросу компании «Пальмира». Задание проекта заключается в том, что необходимо описать требования к системе, которая собирает данные с датчиков ГЛОНАСС/тахографов, учитывает тип автобуса, дорожные условия в 7 районах ЛО, пробки и стиль вождения, чтобы выявлять отклонения от нормативов и предсказывать потребность в ГСМ для закупок. Помимо проработки требований, необходимо разработать пилотную версию системы (MVP), которая будет предсказывать расход топлива на конкретный контракт с учётом ручного ввода маршрутов и сроков планируемых перевозок в систему в соответствии с контрактом. Пилотная версия должна обеспечивать ввод как типовых маршрутов (из утверждённой маршрутной сети), так и разовых — с возможностью задания ключевых параметров (протяжённость, район ЛО, планируемый период). MVP не требует полной автоматизации или глубокой интеграции со всеми системами предприятия, но обязан демонстрировать корректную работу базовой прогнозной модели с учетом описанных выше ограничений.

Компания готова предоставить необходимую поддержку и информацию в объеме, достаточном для выполнения поставленных задач курсовой работы.

Руководителем от компании будет Жукова Марина Анатольевна, генеральный директор ООО «Пальмира».

Проект будет реализован по запросу компании.

Полученные результаты будут использоваться компанией «Пальмира» в своей деятельности.

09.02.2026 г.

Генеральный директор ООО «Пальмира» ***М.А. Жукова***

